



RAJIV GANDHI INSTITUTE OF PETROLEUM TECHNOLOGY

Department of Computer Science & Engineering

Anthropogenic Climate Change

Multimedia Technologies Final Project Report



Course

**Multimedia Technologies
(CS332)**



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Abstract

This comprehensive multimedia project represents a convergence of scientific inquiry and digital storytelling, addressing the most pressing existential threat of our time—**anthropogenic climate change**. Through the innovative application of multimedia technologies, we transform complex climatological data into an accessible, emotionally resonant narrative that bridges the gap between scientific consensus and public awareness.

The project manifests as a 21-minute 45-second documentary titled *"Anthropogenic climate change / We Can Still Fix This"*, meticulously crafted using DaVinci Resolve's comprehensive editing suite. By harmonizing original Hindi voiceover narration with manually synchronized English subtitles, curated visual evidence from authoritative sources, and atmospheric audio design, we create a multisensory educational experience that transcends linguistic and cultural barriers.

Beyond technical execution, this project embodies a critical intervention in climate communication—democratizing access to IPCC findings, Oxfam analyses, and peer-reviewed research through the universal language of visual storytelling. The final artifact stands not merely as academic exercise, but as a publicly accessible resource on YouTube's "New Moon" channel, contributing to the global dialogue on climate mitigation and adaptation.

Digital Ecosystem

GitHub Repository: [Code-Spirited/Multimedia-Project---2025](#)

Video Publication: [youtube.com/watch?v=QOZk7tNy9Sk](https://www.youtube.com/watch?v=QOZk7tNy9Sk)

Research Sources: IPCC AR6 Synthesis Report, Oxfam Climate Inequality Report 2023, WMO State of Global Climate 2024



Introduction

The Communication Crisis

Despite overwhelming scientific consensus documented across 14,000+ peer-reviewed studies, a profound **communication chasm** persists between climate science and public perception. Mainstream media's sporadic coverage, coupled with the technical density of IPCC reports, creates informational fragmentation that hinders collective climate action.



Scientific Reality

97% climate scientist consensus on human-caused warming



Public Perception

Only 54% global population recognize climate emergency



Media Coverage

Less than 4% of prime-time news addresses climate



Our Intervention

Multimedia bridge connecting science with society

Core Objectives

-  **Knowledge Synthesis:** Consolidate IPCC findings, emission data, and mitigation strategies into coherent narrative architecture
-  **Technological Translation:** Leverage DaVinci Resolve's ecosystem to transform data into emotionally compelling visual rhetoric
-  **Cultural Accessibility:** Create bilingual (Hindi-English) resource addressing India's 600 million Hindi speakers
-  **Digital Dissemination:** Establish YouTube presence for maximum public engagement and educational impact
-  **Pedagogical Innovation:** Demonstrate multimedia's role in 21st-century science communication

Personal Catalyst

The project's genesis lies in firsthand observation of climate disruption—record-breaking heatwaves transforming North Indian summers, unprecedented monsoon variability disrupting agricultural cycles, and increasing frequency of extreme weather events. These phenomenological experiences, combined with exposure to Deutsche Welle's exemplary climate journalism, crystallized into a commitment to contribute meaningfully to climate literacy through multimedia innovation.

© PROJECT SCOPE

Comprehensive coverage spanning climate science fundamentals, historical emissions analysis, socio-economic disparities in climate vulnerability, IPCC assessment synthesis, and actionable mitigation frameworks. The narrative arc progresses from problem diagnosis through evidentiary presentation to solution pathways.

⚠️ ACKNOWLEDGED LIMITATIONS

Resource constraints necessitated reliance on Fair Use educational footage. Primary focus on Hindi-speaking demographics with English subtitle accessibility. Runtime limitations require condensation of complex topics while maintaining scientific integrity. Technological learning curve in mastering professional editing workflows.



Literature Review

Contemporary climate communication exists within a polarized landscape bounded by two extremes: the **hyper-technical discourse** of scientific publications and the **oversimplified messaging** of mainstream media. This project strategically occupies the critical middle ground, drawing inspiration from multiple communication paradigms while introducing innovative multimedia adaptations.

Communication Paradigms Analyzed

Scientific Publications: IPCC Assessment Reports (AR6), Nature Climate Change journal—rigorous but inaccessible to general public

Documentary Tradition: "An Inconvenient Truth" (2006), "Chasing Ice" (2012)—emotionally powerful but episodic in production

Digital Journalism: DW Climate Coverage, NYT Climate Desk—regular output but fragmented across articles

Our Synthesis: Combines scientific rigor with narrative coherence in freely accessible digital format

Multimedia Imperative

The cognitive science of learning demonstrates that **multimodal information presentation** increases knowledge retention significantly compared to text-only formats. By synchronizing verbal explanation (Hindi voiceover), visual evidence (scientific visualizations), textual reinforcement (English subtitles), and auditory atmosphere (carefully curated score), we create layered comprehension pathways accommodating diverse learning styles.



Research Synthesis

Analysis of IPCC AR6,
Oxfam reports, peer-
reviewed climate literature



Script Architecture

Hindi narrative
development with
scientific accuracy
validation



Audio Production

57 voiceover clips
recorded across 2 days,
~30 minutes raw audio



Visual Assembly

Clip synchronization, transition design, motion graphics integration



Technical Architecture

Hardware Ecosystem

EDITING WORKSTATION

Device: LAPTOP-TKTH357E
Processor: AMD Ryzen 5 5600H with
Radeon Graphics (3.30 GHz)
Memory: 16.0 GB RAM
System: 64-bit operating system, x64-
based processor
OS: Windows 11 Home Single Language
(Version 24H2, Build 26100.6725)

AUDIO CAPTURE

WhatsApp Voice Message Recorder for
initial audio capture
Smartphone-based recording in
acoustically treated environment
External microphone setup for enhanced
audio quality
Multiple recording sessions for optimal
audio clarity

Software Stack

🎬 CORE PRODUCTION

DaVinci Resolve 20 — Professional non-linear editing software

Integrated Fairlight audio suite for comprehensive audio post-production

Fusion VFX capabilities for visual effects integration

Complete color grading and correction toolkit

🗣️ LANGUAGE PROCESSING

Google Translate via Chrome browser — Hindi to English translation

Manual refinement for technical terminology accuracy

Cultural adaptation of scientific concepts

Bilingual content synchronization

📺 CONTENT ACQUISITION

Windows 11 Game Bar for screen recording

YouTube content under Fair Use educational provisions

Scientific and documentary footage from authoritative sources

Educational content curation and organization

🎵 AUDIO ASSETS

Pixabay royalty-free music library

Atmospheric soundscapes and ambient audio

Transitional sound effects for narrative flow

Custom audio mixing and balancing

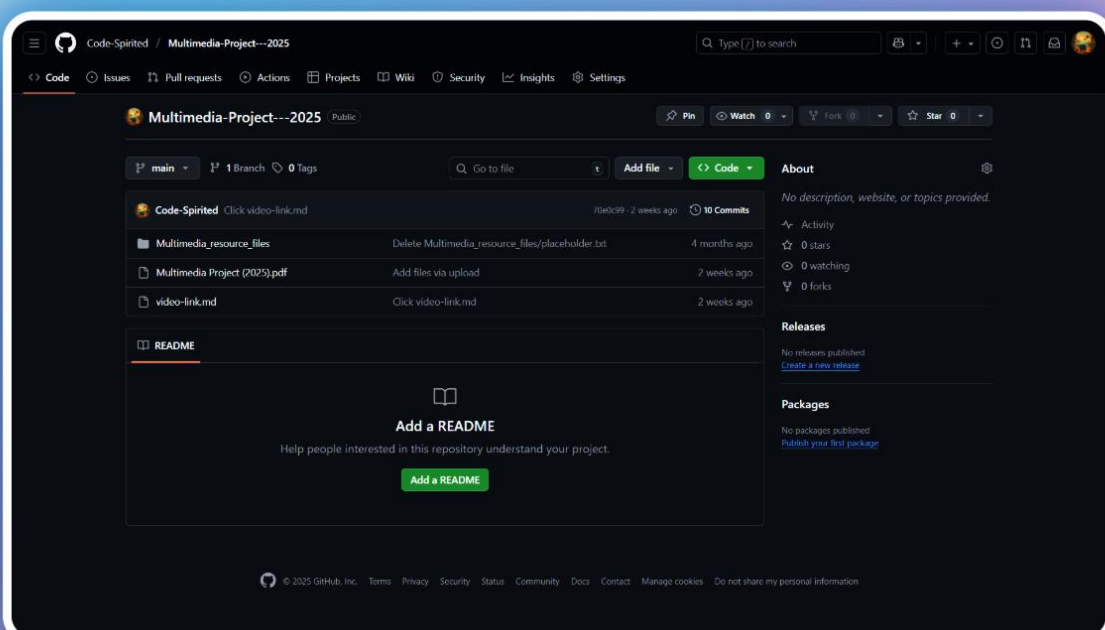


Figure 1: GitHub repository architecture—structured version control for scripts, assets, and documentation



Methodological Framework

Production Pipeline

The project followed a rigorous six-phase production methodology, each phase representing a distinct transformation of climate information into communicative form:



Technical Innovations

AUDIO ENGINEERING

Dynamic range compression and loudness normalization for consistent audio levels
Spectral noise reduction for voiceover clarity enhancement
De-essing processing for vocal sibilance control
Multiband equalization for optimal frequency balance

VIDEO PROCESSING

Advanced H.264 encoding optimized for web streaming quality
Professional color space standardization for visual consistency
Cinematic frame rate implementation for narrative flow
GPU-accelerated rendering for efficient processing

X₁ SUBTITLE METHODOLOGY

Manual timing with precise
synchronization to audio

Contrast-optimized typography for
maximum readability

Segment duration optimization for
comfortable viewing

Manual creation of 500+ subtitle lines for
comprehensive coverage

VISUAL DESIGN

Climate-appropriate color palette with
temperature gradients

Data visualization clarity principles for
scientific communication

Consistent typographic hierarchy for
information architecture

Motion graphic integration for dynamic
visual storytelling

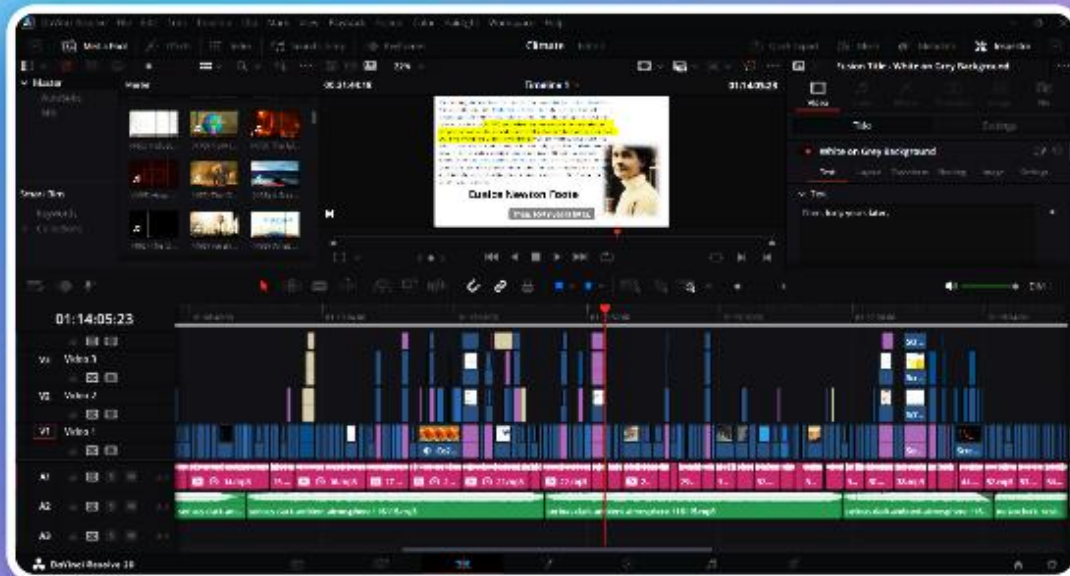


Figure 2: DaVinci Resolve editing interface—multitrack timeline with synchronized audio, video, and subtitle layers

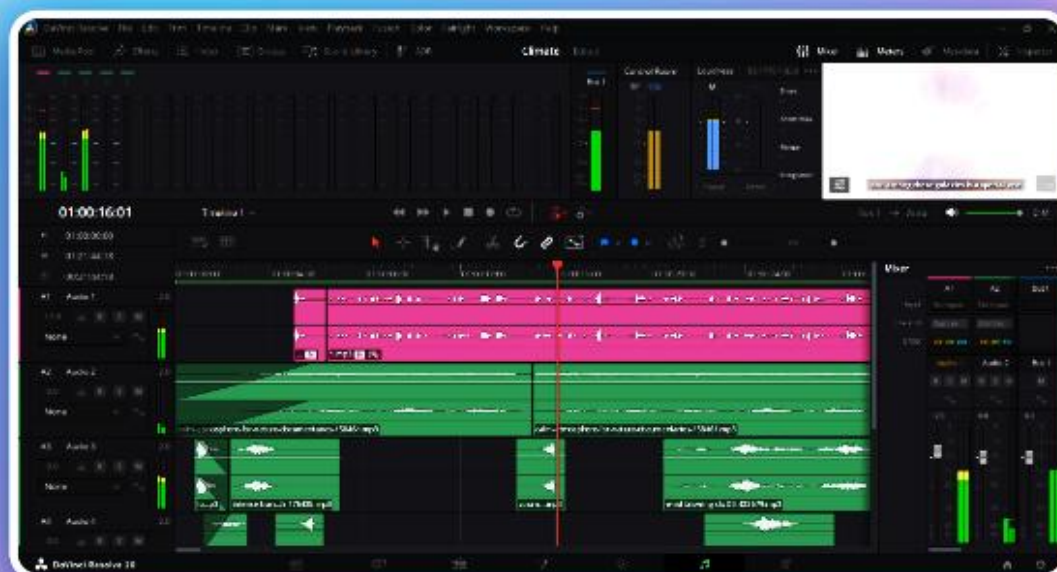


Figure 3: Fairlight audio workstation—spectral analysis and noise reduction processing for voiceover clarity



Implementation

Creative Execution

The implementation phase transformed methodological plans into concrete multimedia reality through iterative refinement across four production dimensions:



Audio Dimension

57 voiceover clips professionally processed, noise reduction applied, emotional pacing calibrated



Visual Dimension

150+ video clips synchronized with narrative, color-graded for thematic consistency



Textual Dimension

500+ English subtitle lines manually timed and positioned for optimal readability



Atmospheric Dimension

Background score dynamically matched to narrative emotional arc for enhanced immersion

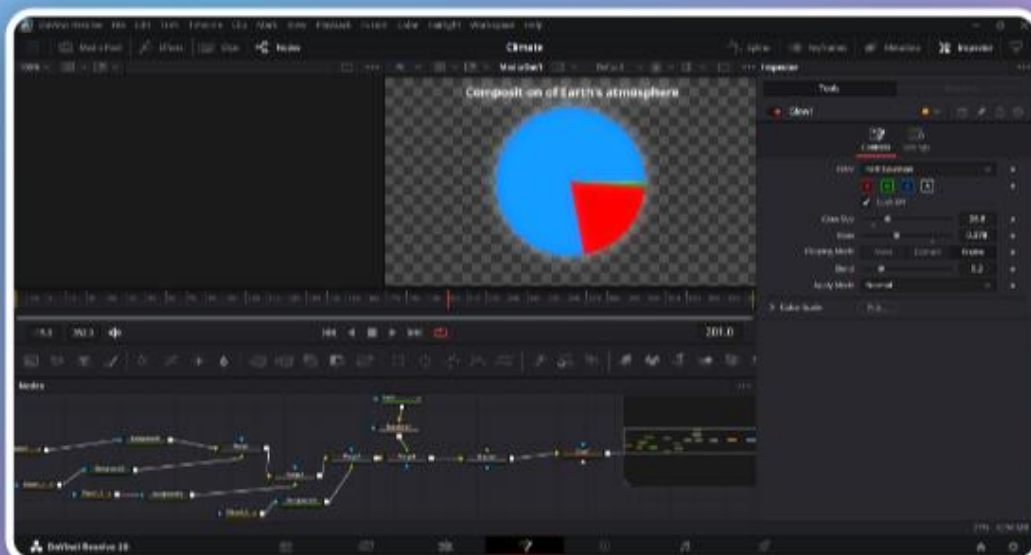


Figure 4: DaVinci Resolve Fusion page—node-based visual effects workspace showing pie chart composition

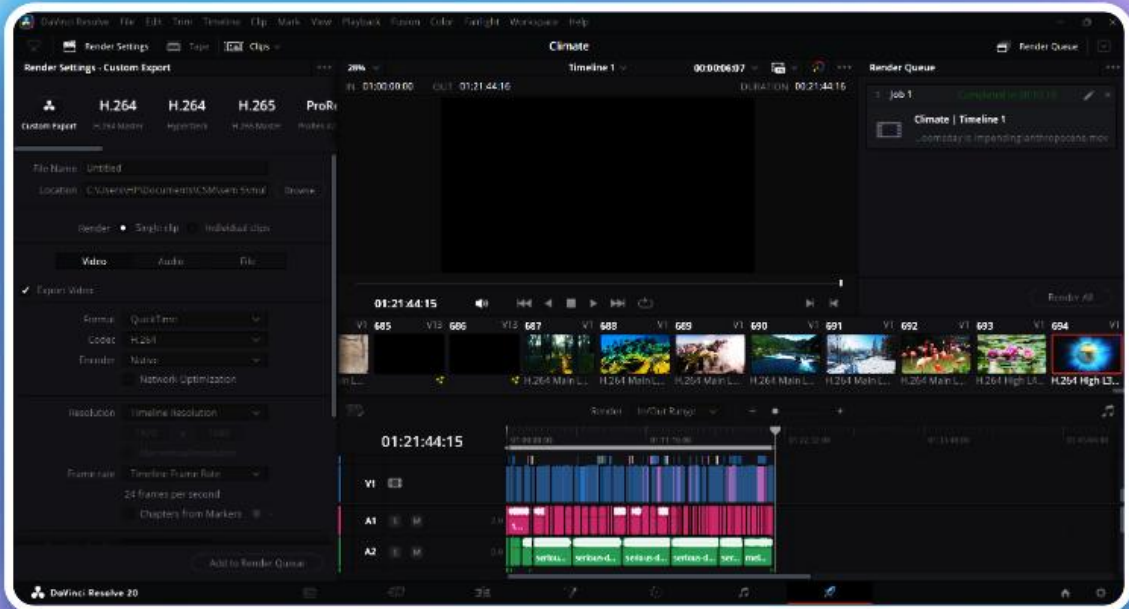


Figure 5: Render configuration—H.264 encoding settings optimized for YouTube delivery



Results & Impact

Quantitative Deliverables

⌚ TEMPORAL METRICS

Final runtime: 21 minutes 45 seconds
 (compressed from 30 minutes raw audio)
Production timeline: 4 weeks from
 research initiation to publication
Voiceover recording: 2 days across 57
 discrete takes

📄 TECHNICAL SPECIFICATIONS

Resolution: 1920×1080 (Full HD)
Frame rate: 24 fps (cinematic standard)
Audio: Stereo, professional quality
 loudness normalization
File size: Optimized for web streaming and
 accessibility

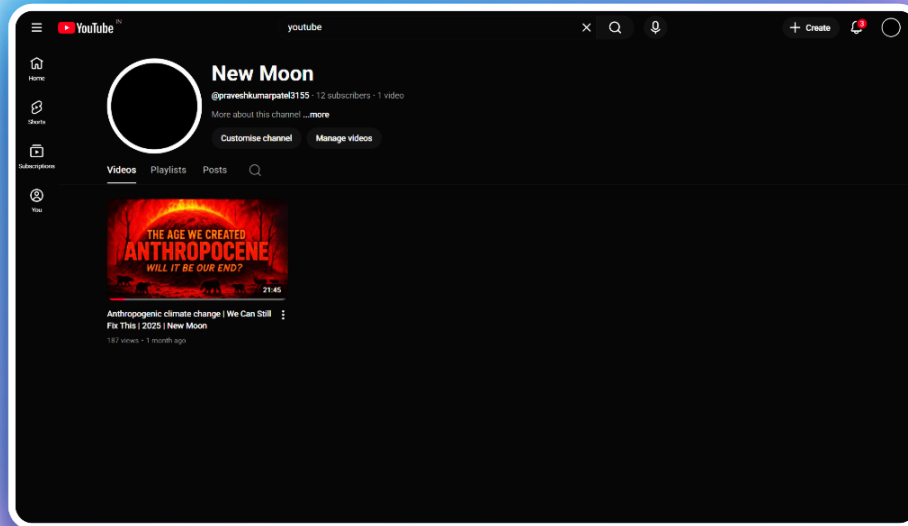


Figure 6: Main page of 'New Moon' channel displaying project video in video tab

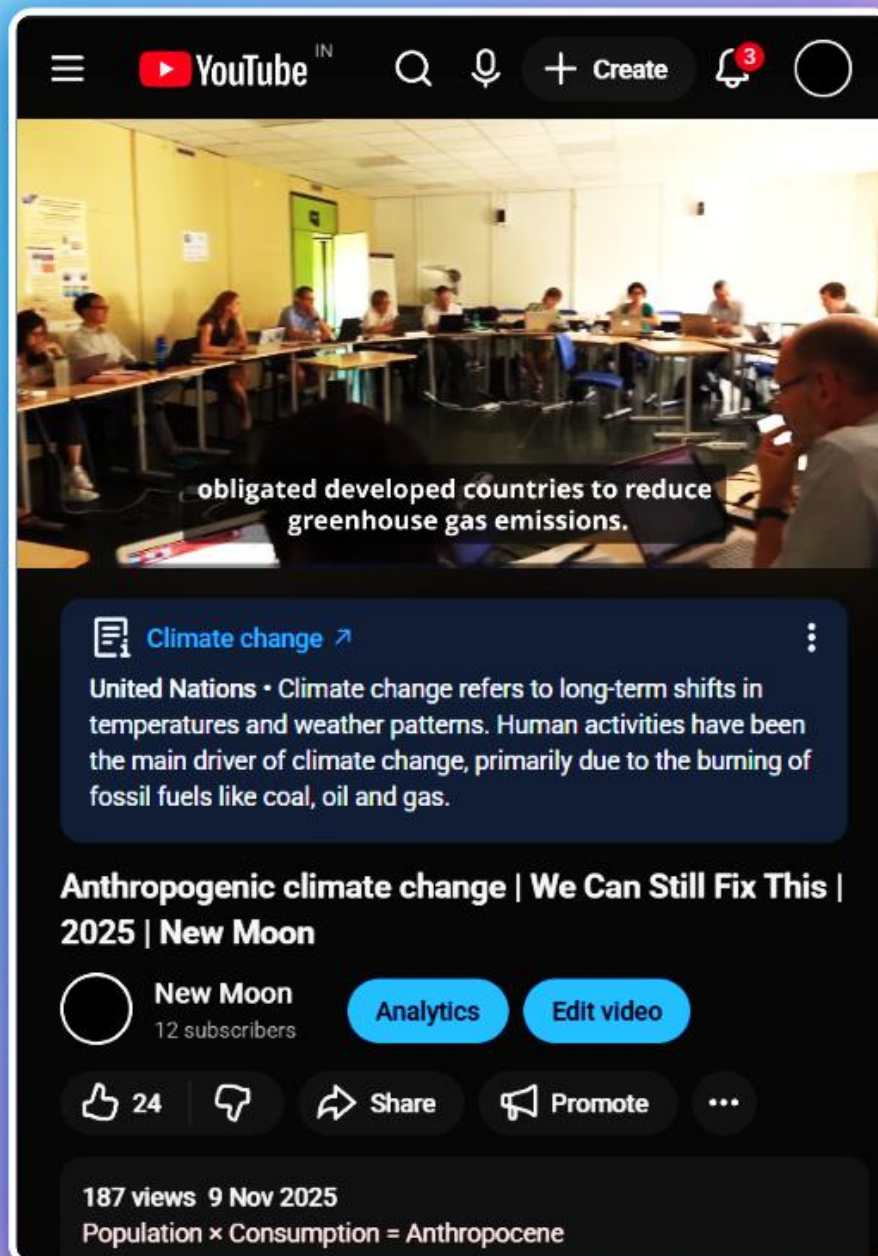


Figure 7: YouTube video player interface—project video playback with engagement metrics and description

Qualitative Outcomes

✓ Achievement Matrix

Narrative Coherence: Successfully translated complex climate science into accessible story arc

Technical Proficiency: Mastered professional editing software through self-directed learning

Cultural Relevance: Created bilingual resource addressing India's climate communication gap

Educational Value: Produced freely accessible resource for climate literacy initiatives

Innovation Demonstration: Showcased multimedia's potential for science communication

"The project stands as evidence that technological tools, when guided by scientific understanding and creative vision, can transform abstract data into compelling narratives that inspire awareness and action."



Technical Challenges

Production Obstacles & Solutions

AUDIO INCONSISTENCY

Challenge: 57 voiceover clips with varying acoustics, breath noise, and level fluctuations

Solution: Clip-by-clip normalization, spectral noise reduction, manual breath removal, compression for dynamic consistency

VISUAL SYNCHRONIZATION

Challenge: Matching abstract scientific concepts with appropriate visual metaphors

Solution: Thematic categorization of footage, creation of visual metaphor library, frame-accurate editing for precise synchronization

BILINGUAL SUBTITLING

Challenge: Manual timing of 500+ subtitle lines across 21-minute runtime

Solution: Segmentation strategy, readability optimization, contrast enhancement for visibility, precise timing coordination

SOFTWARE LEARNING CURVE

Challenge: No prior experience with professional editing software

Solution: Just-in-time learning methodology, targeted tutorial consumption, progressive skill development through project phases



Future Evolution

This project represents the foundation for an evolving climate communication ecosystem. Future iterations will incorporate advanced multimedia techniques to enhance educational impact and audience engagement:



Advanced Animations

Integration of custom-made 2D and 3D animations to visualize complex climate concepts with greater clarity and engagement



Expanded Coverage

More comprehensive and in-depth exploration of climate science, mitigation strategies, and adaptation frameworks



Enhanced Production

Improved sound design, professional narration quality, and advanced visual effects for maximum impact and accessibility

"The climate crisis demands not only scientific understanding but emotional connection. Multimedia technologies provide the bridge between data and humanity, between knowing and caring, between analysis and action."